

=> Access specifier :- { public, protected, default, private }

=> Package :-
⇕

Project :- 100 files, 200 files

Folder
⇕
files

500 files

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=> Banking Project => Register login

=> web layer service DAO

Java Project

↳ web layer
↳ service
↳ DAO

lang util sql net
⇕
package

Java

↳ journey - - - - - string

import java.util.Scanner;

↳ util -> Scanner, Array

↳ sql

↳ net

≡

→ com.telusko.login

↳ ≡ 2 files

→ com.telusko.register

↳ ≡ 3 files

→ com.telusko.main

↳ ≡

main

import com.telusko.*;

main

≡

public ☐
 protected ☐
 default ☐
 private ☐

}
 package ☐

variable $\Rightarrow$ Data to be stored

{ int age = 18; }

Array $\Rightarrow$ large Data $\Rightarrow$ similar / Homogeneous

↳ Limitations $\Rightarrow$ only Homogeneous

$\Rightarrow$ fixed size (cannot grow / shrink)

$\Rightarrow$ contiguous location.

$\Rightarrow$ Arrays $\rightarrow$ utility

~~Legacy~~
 $\Rightarrow$ Vector
 $\equiv$
 Dictionary
 $\equiv$
 stack

open source

Org $\rightarrow$ Alpha

Java 1.0

Person

Person

Person

collection

Java 1.2

Java 2

$\Rightarrow$ Joshua $\Rightarrow$

interface & 7 classes

↑↑

Collection $\rightarrow$ Java 2

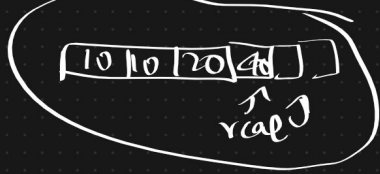
{
 ↳ setg interface & classes & each class
 with specific Data structure & algorithm
 to handle & manage large volume of Data

both homogenous & heterogeneous

ArrayList = y DS → Dynamic Array DS

↳ Size can grow or shrink
Dynamically

list.add(40);



↳ can store Both homogenous & heterogeneous type Data

↳ Data is stored as object.

→ implement List interface.

→ indexed Based access is allowed.

al.add(100);
al.add(200);
al.add(300);

al.add(100, 200, 300) real

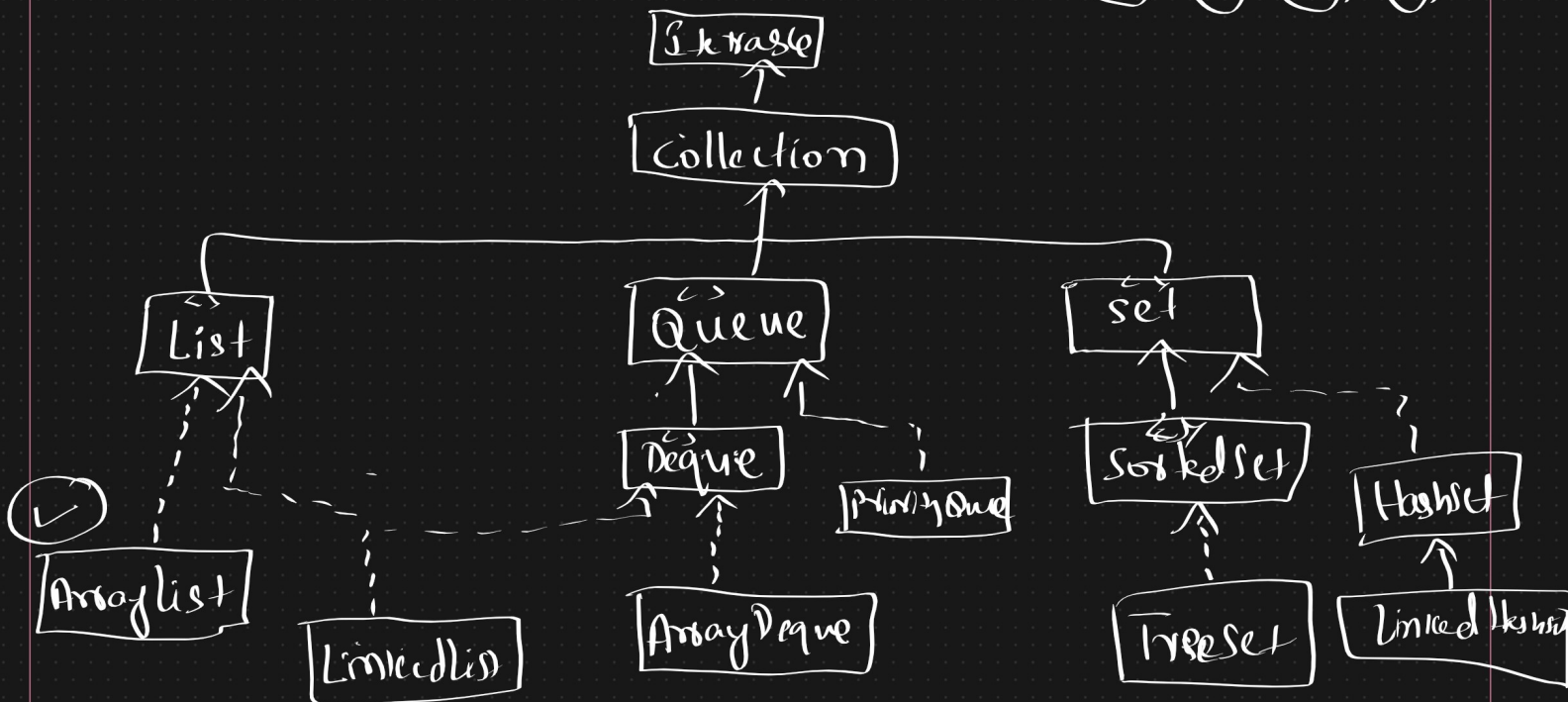
al.add(1, 600);

al.add(0, 500);

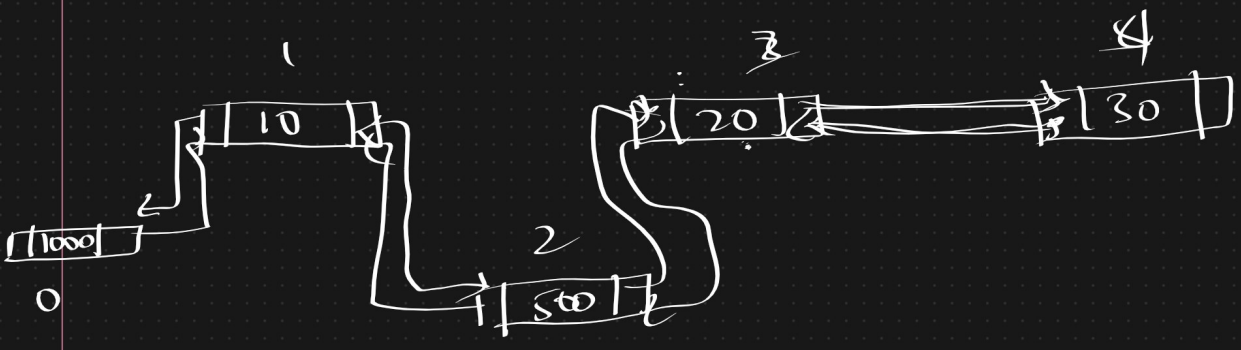
Diagram showing array state after adding 100, 200, 300. The array contains [100, 600, 200, 300] with indices 0, 1, 2, 3, 4 below it.

0.1 Ms

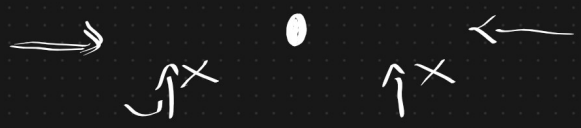
Diagram showing array state after adding 500, 100, 600, 200, 300. The array contains [500, 100, 600, 200, 300] with indices 0, 1, 2, 3, 4 below it.



Doubly linked list :- `ll.add(2, 500)`



ArrayDeque :- (Double ended Queue)



=> rear end & front end



```

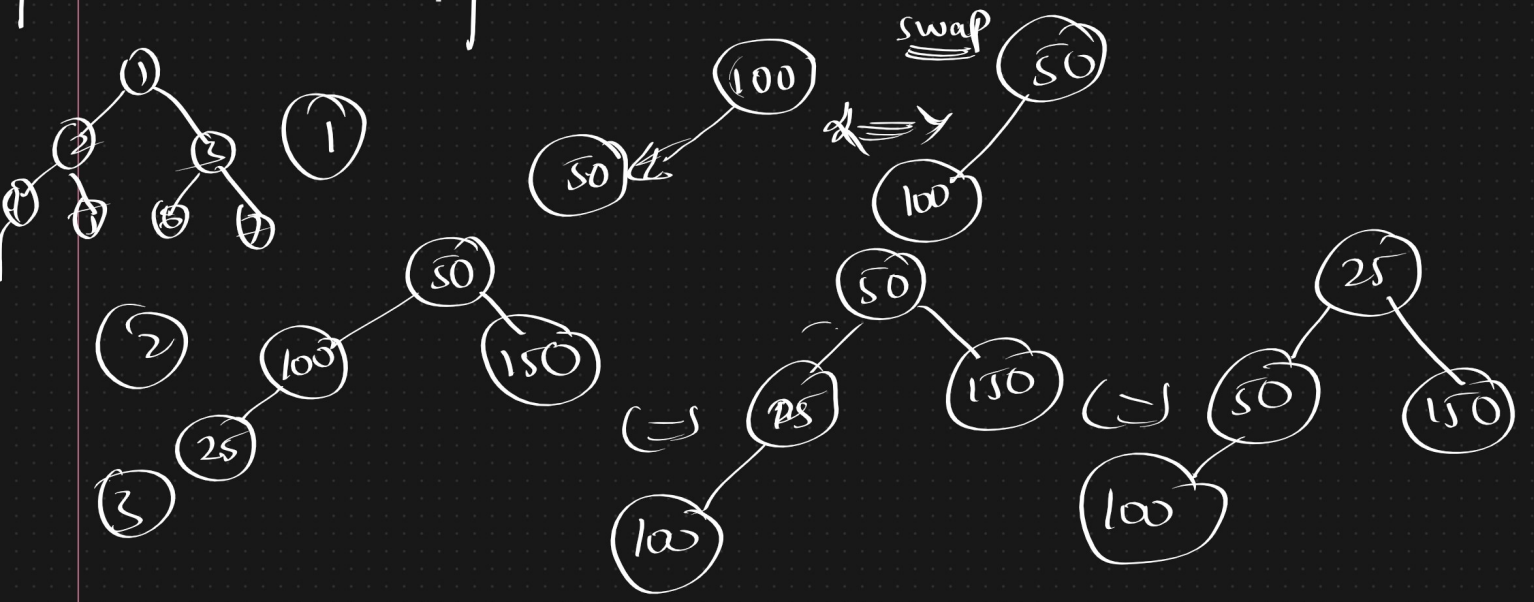
pq.add(100);
pq.add(50);
pq.add(150);
pq.add(25);
pq.add(75);
pq.add(125);
pq.add(175);

```

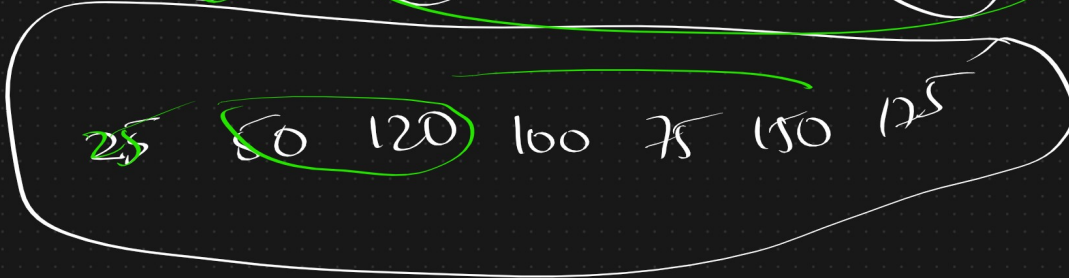
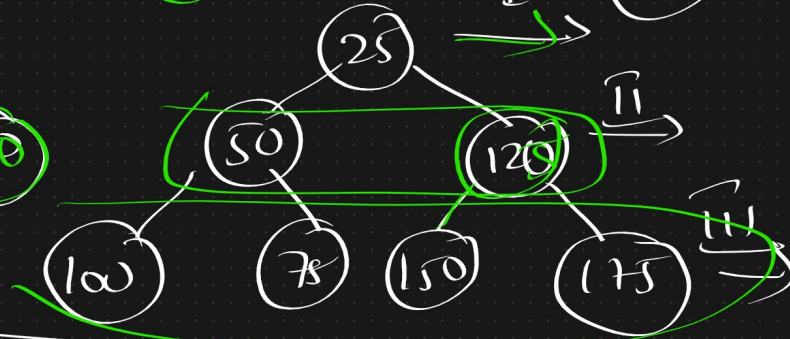
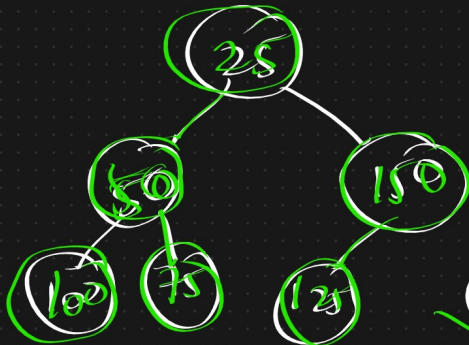
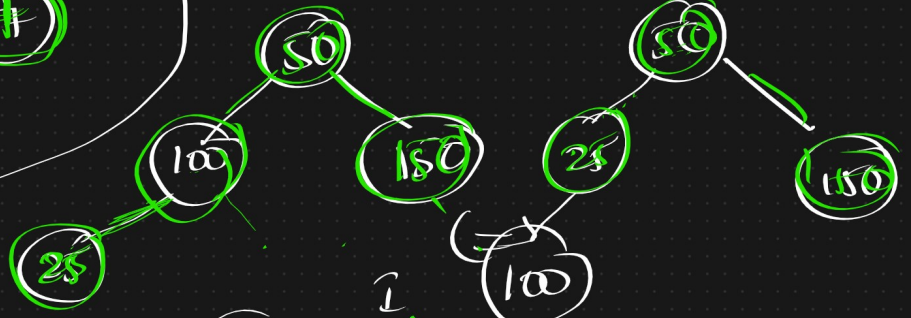
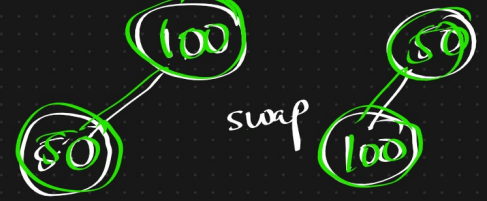
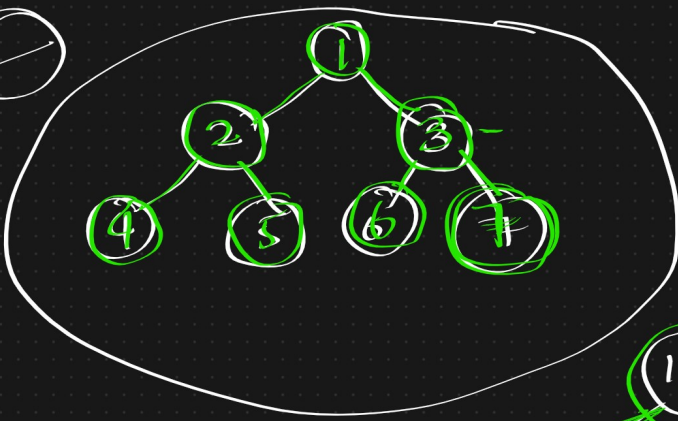
100 50 150 25 75 125 175

min-Heap :-

[25, 50, 125, 100, 75, 150, 175]



①



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